Reproducing a memory benchmark result, then testing where it breaks.

Does write-side state adjudication survive small models?

Reproducing and stress-testing STALE and CUPMem

1 The problem

- Assistants keep long-term memories about the user
- New evidence can make an old memory false **without contradicting it**
- The paper calls this an **implicit conflict**

Agent memories can go out of date without anyone saying so.

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2 STALE benchmark

- 400 scenarios, 1,200 probes, ~150K-token histories
- Probes: **state resolution**, **premise resistance**, **policy adaptation**
- Best frontier model: **55.2%** overall

Finding the update is not the same as acting on it.

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3 CUPMem

- Decide what is still true **at write time**
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The proposed fix: adjudicate state when memory is written.

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Does the fix still work when the adjudicator is small and cheap?

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Reproducing and stress-testing STALE and CUPMem

4 Our questions

- **RQ1** Does 68% reproduce?
- **RQ2** Does it hold with small 3–9B adjudicators?
- **RQ3** Which stage fails? **RQ4** What does each part cost?
- **RQ5** Do 2-hop cascades break it?

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Same data, same judge, same prompts, so differences mean something.

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5 Approach

- Three-way check: **paper vs official code vs our reimplementation**
- Baselines: full context, naive RAG, Mem0
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Which setting produced 68% may matter as much as the method.

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- GPT-4o-mini accepts false premises, matching the paper

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Reproducing and stress-testing STALE and CUPMem

Reproduction first, then the stress tests, written up by December.

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Roadmap

- 13-week plan, started 28 Sep 2026
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Can a cheap model reliably decide which memories are still true?

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